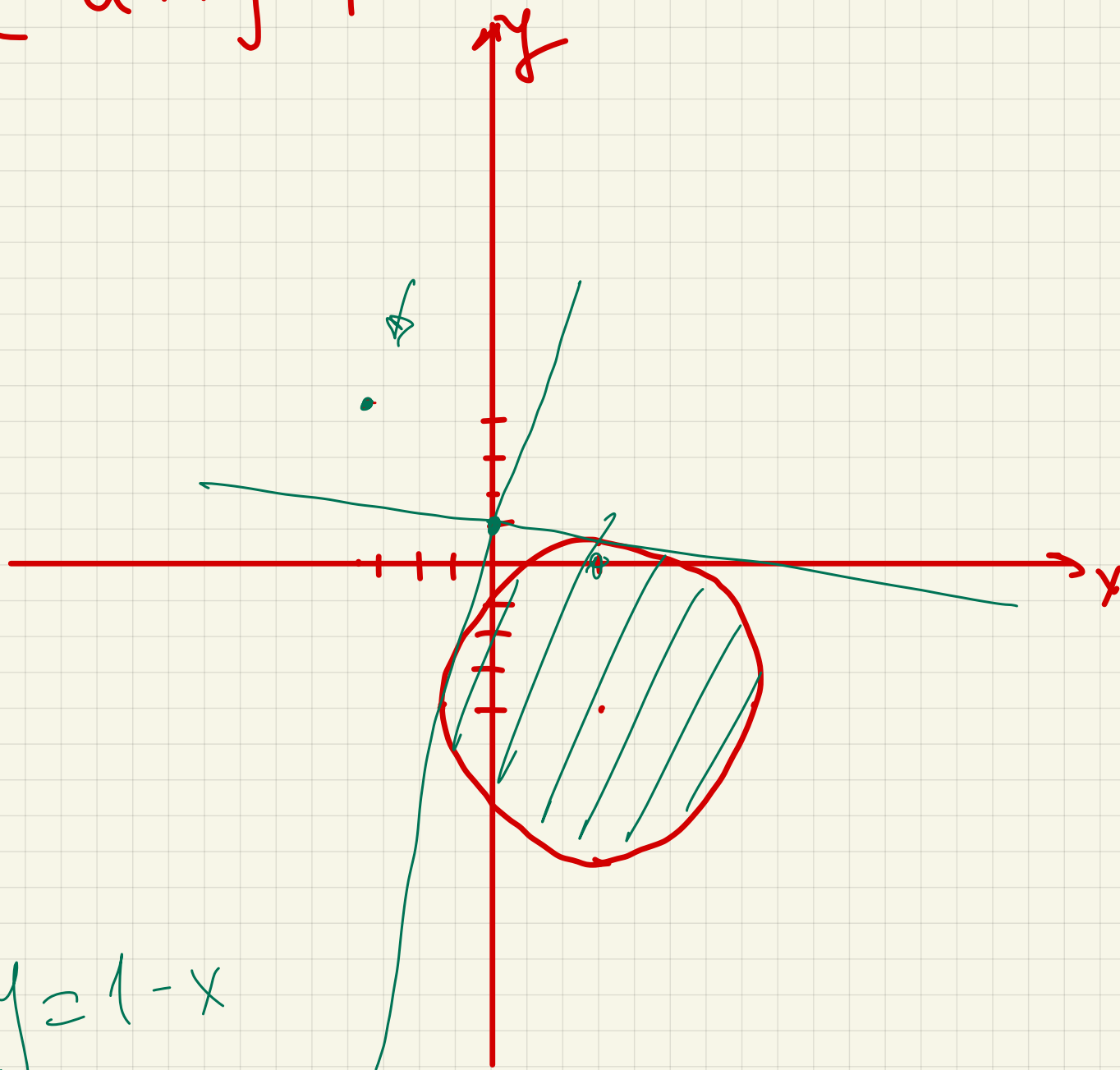




$$\begin{cases} ((x-3)^2 + (y+4)^2 - 17)((2x+7)^2 + (2y-9)^2) \leq 0 \\ ax + y = 1 \end{cases}$$



$$y = 1 - x$$

$$y = 1 - ax$$

$$\frac{9}{2} = 1 - a \cdot \left(-\frac{7}{2}\right)$$

$$\frac{9}{2} = 1 + \frac{7}{2}a$$

$$\frac{\frac{2}{2}}{2} = \frac{\frac{4}{2}}{2} a$$

$$a = 1$$

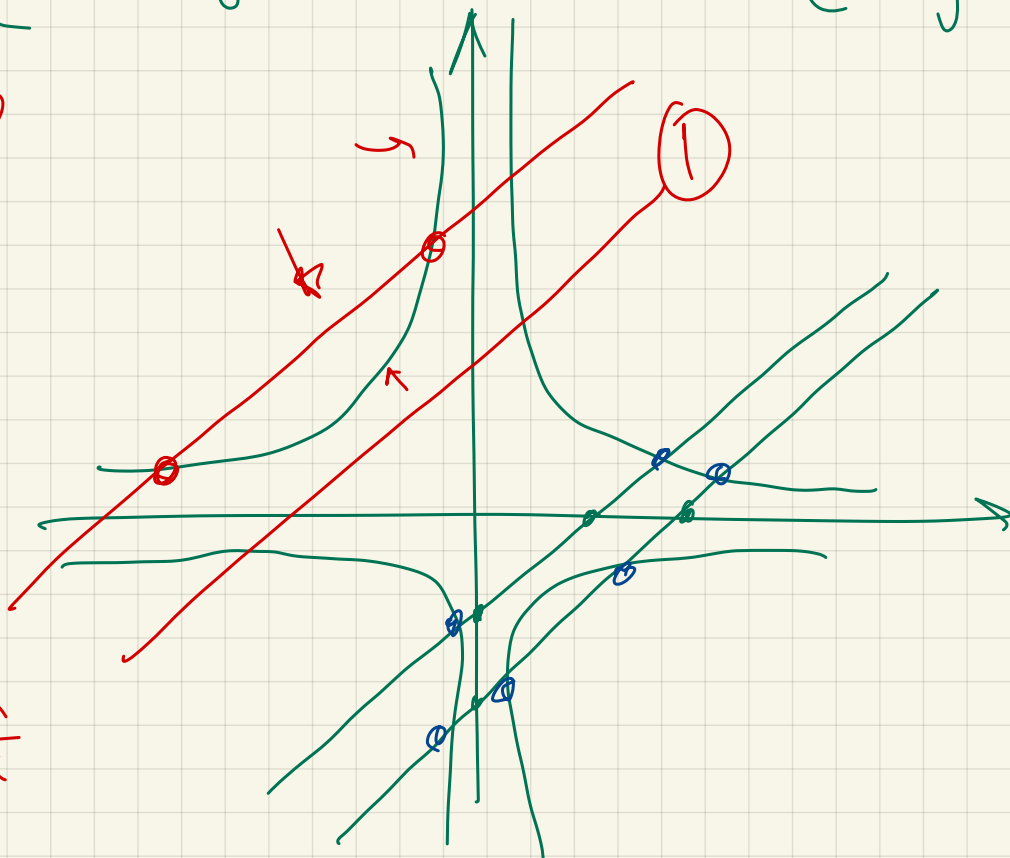
?!.

$$\begin{cases} (ay - ax + 2)(y - x + 3a) = 0 \\ |xy| = a \end{cases}$$

$$\begin{cases} \begin{cases} ay - ax + 2 = 0 \\ y - x + 3a = 0 \end{cases} \rightarrow y = \frac{ax + 2}{a} = \\ \begin{cases} y = x - \frac{2}{a} \\ y = x - 3a \end{cases} \\ xy = \pm a \end{cases}$$

$$a < 0 \\ a = -1$$

$$y = -\frac{a}{x}$$

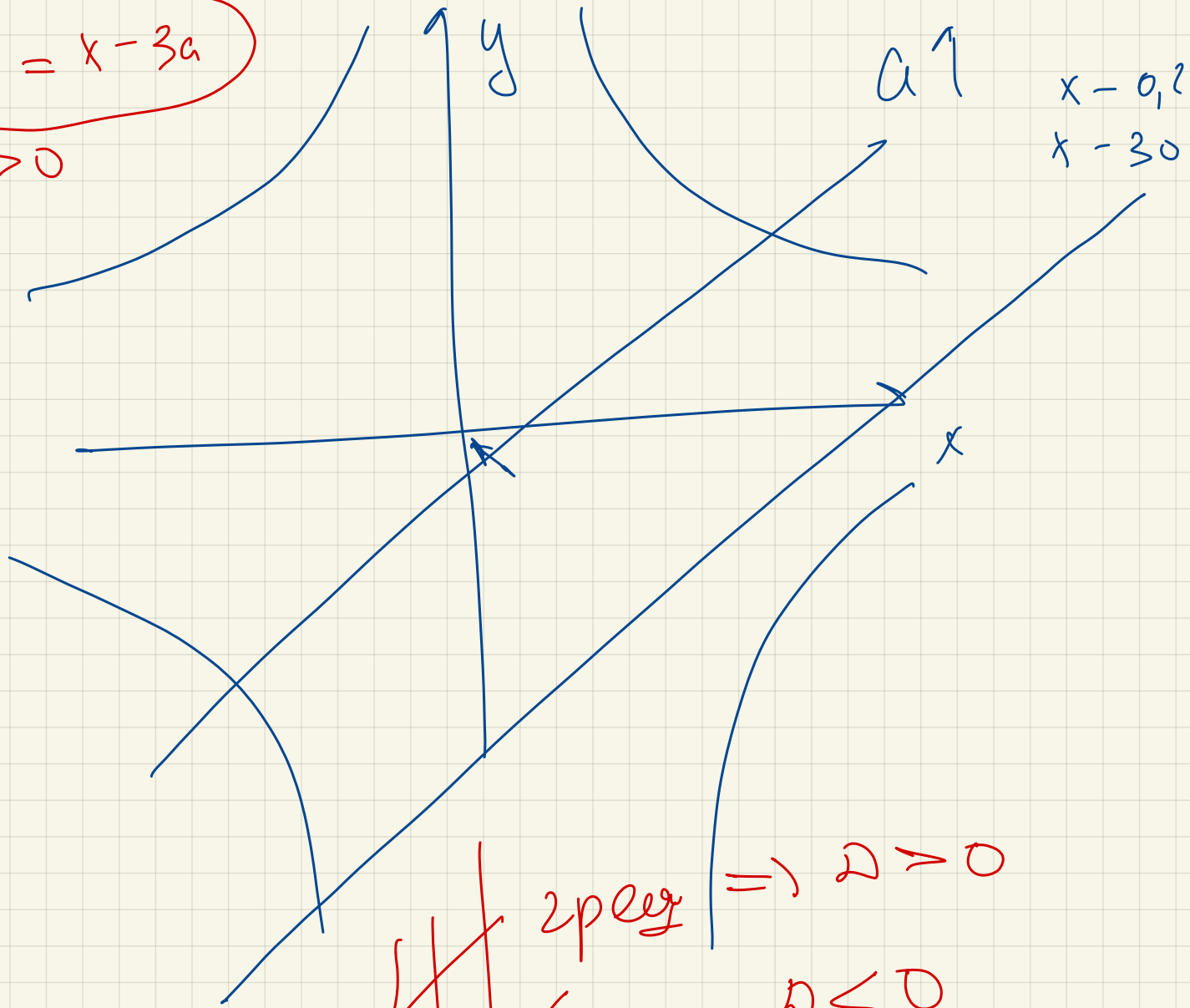


$$\begin{cases} \begin{cases} x - 2 \\ x - 3 \end{cases} \\ y = \frac{a}{x} \\ y = -\frac{a}{x} \end{cases}$$

$$a = 1$$

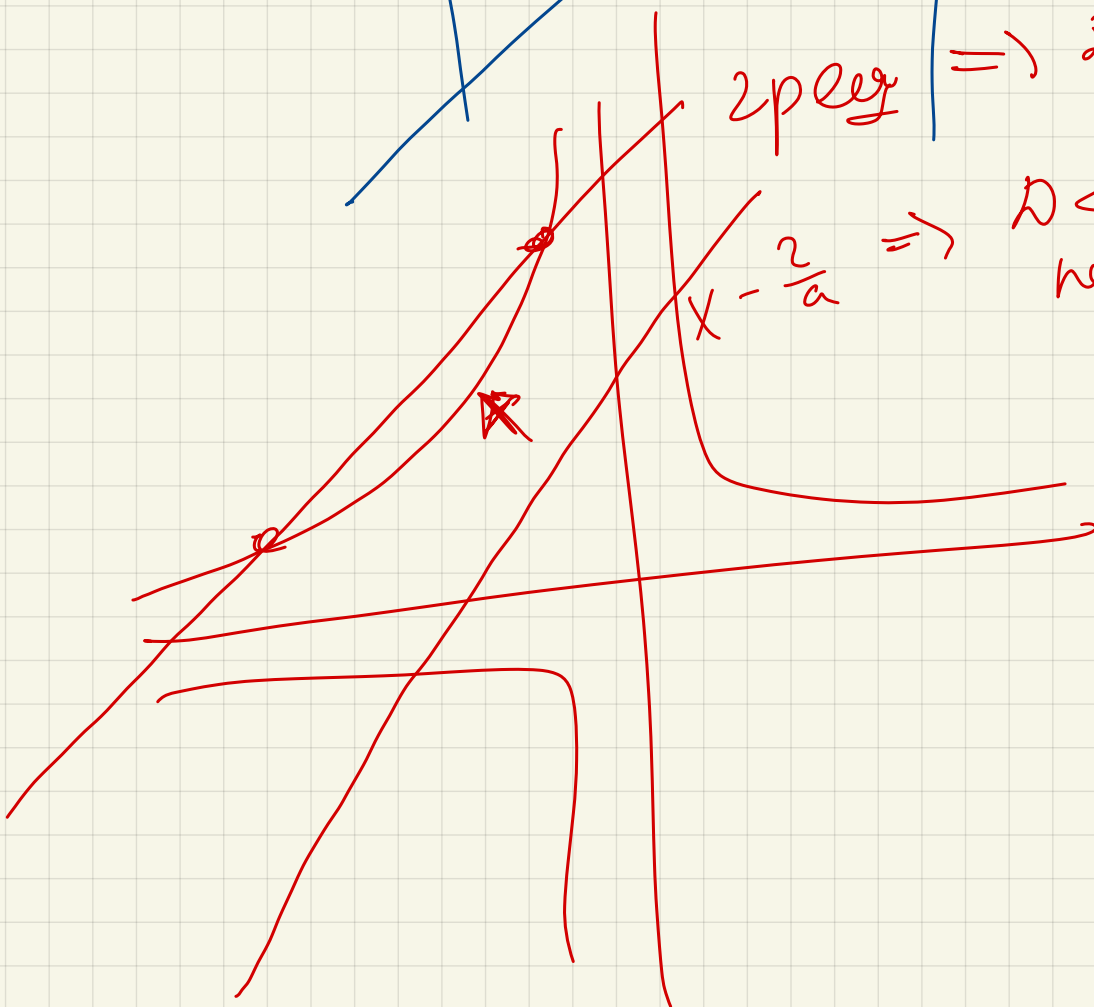
$$\frac{-a}{x} = x - 3a$$

$D > 0$



2 пересечения $\Rightarrow D > 0$

$x - \frac{2}{a} \Rightarrow D < 0$
пересечений



$$\begin{cases} x - 3a = -\frac{a}{x} \\ x - \frac{2}{a} = -\frac{a}{x} \end{cases}$$

$$a > 0$$

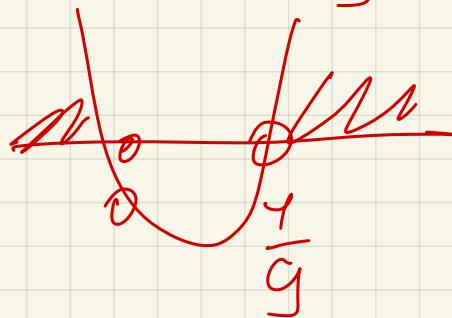
$$a < 0$$

$$x^2 - 3ax + a = 0$$

$$a = 0$$

$$a = \frac{4}{9}$$

$$9a^2 - 4a > 0$$



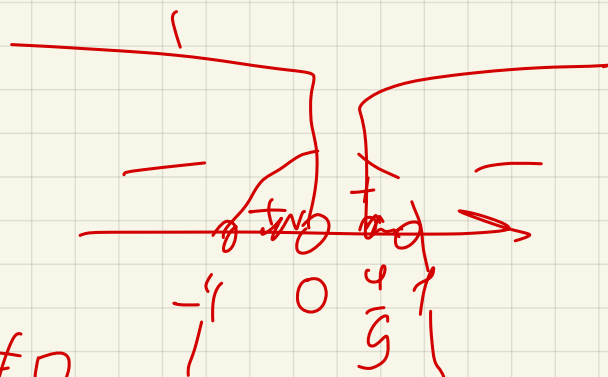
$$x^2 - \frac{2}{a}x + a = 0$$

$$\frac{4}{a^2} - 4a > 0$$

$$\frac{4 - 4a^3}{a^2} > 0$$

$$a \neq 0$$

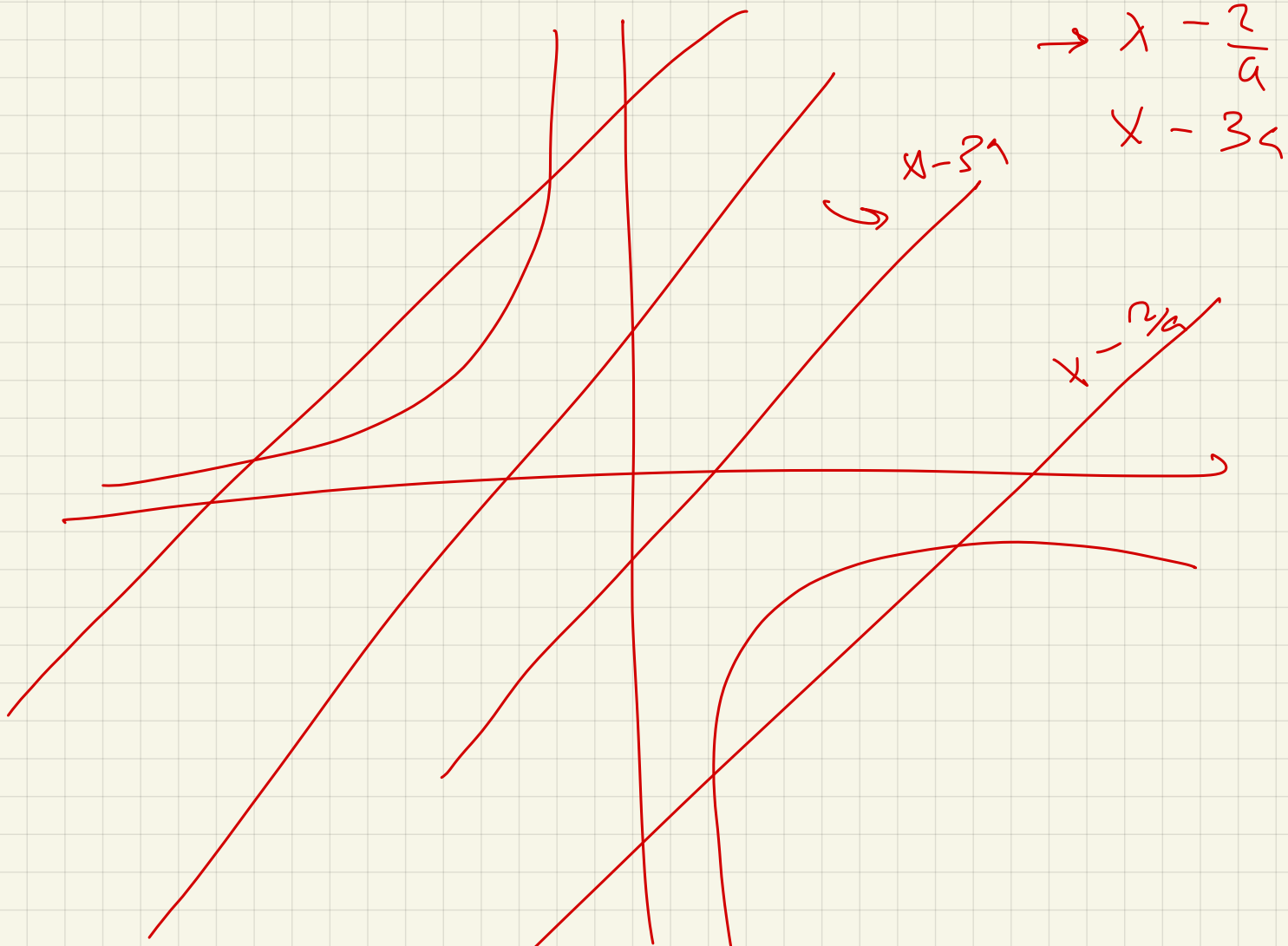
$$a = \pm 1$$



$$a \in (-1; 0) \cup (0; 1)$$

$$a \in (-1; 0) \cup (0; 1)$$

$$0 < a < 1$$



$$\rightarrow x - \frac{2}{a}$$

$$x - 3a$$

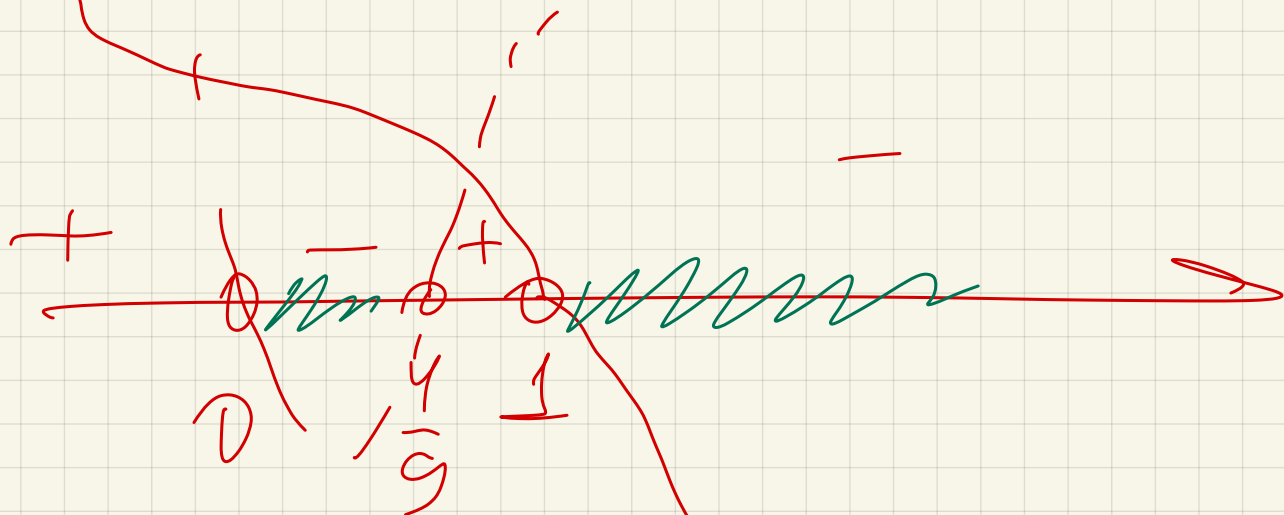
$$-\frac{a}{x} = x - \frac{2}{a} \quad \Delta > 0$$

$$-\frac{a}{x} = x - 3a \quad \Delta < 0$$

$$\Delta_1$$

$$\Delta_2$$

$$\left(\frac{4}{a^2} - 4a \right) (9a^2 - 4a) < 0$$



$$a \in (0; \frac{4}{9}) \cup (1; +\infty)$$

$$\cancel{X-3a} \neq \cancel{X-\frac{2}{a}} \quad \cancel{X \leq 0}$$

$$a^2 = \frac{2}{3} \quad a \neq \pm \sqrt{\frac{2}{3}}$$

$$\left[\begin{array}{l} \left\{ \begin{array}{l} \log_2 (x^2 + 2ax + a^2 - a + 1) \leq 0 \\ 2 - \log_2 \frac{a^2}{6} \geq 0 \end{array} \right. \\ \\ \left. \begin{array}{l} \geq 0 \\ \leq 0 \end{array} \right. \end{array} \right.$$

$$\begin{cases} x^2 + 2ax + a^2 - a + 1 \leq 2^0 - 1 \\ \frac{a^2}{6} \leq 4 \leftarrow a^2 \leq \sqrt{24} \\ -\sqrt{24} \leq a \leq \sqrt{24} \\ x^2 + 2ax + a^2 - a + 1 > 0 \\ \frac{a^2}{6} \geq 4 \rightarrow a^2 \geq 24 \\ -\sqrt{24} \geq a \geq \sqrt{24} \end{cases}$$

$$x^2 + 2ax + a^2 - a \leq 0$$

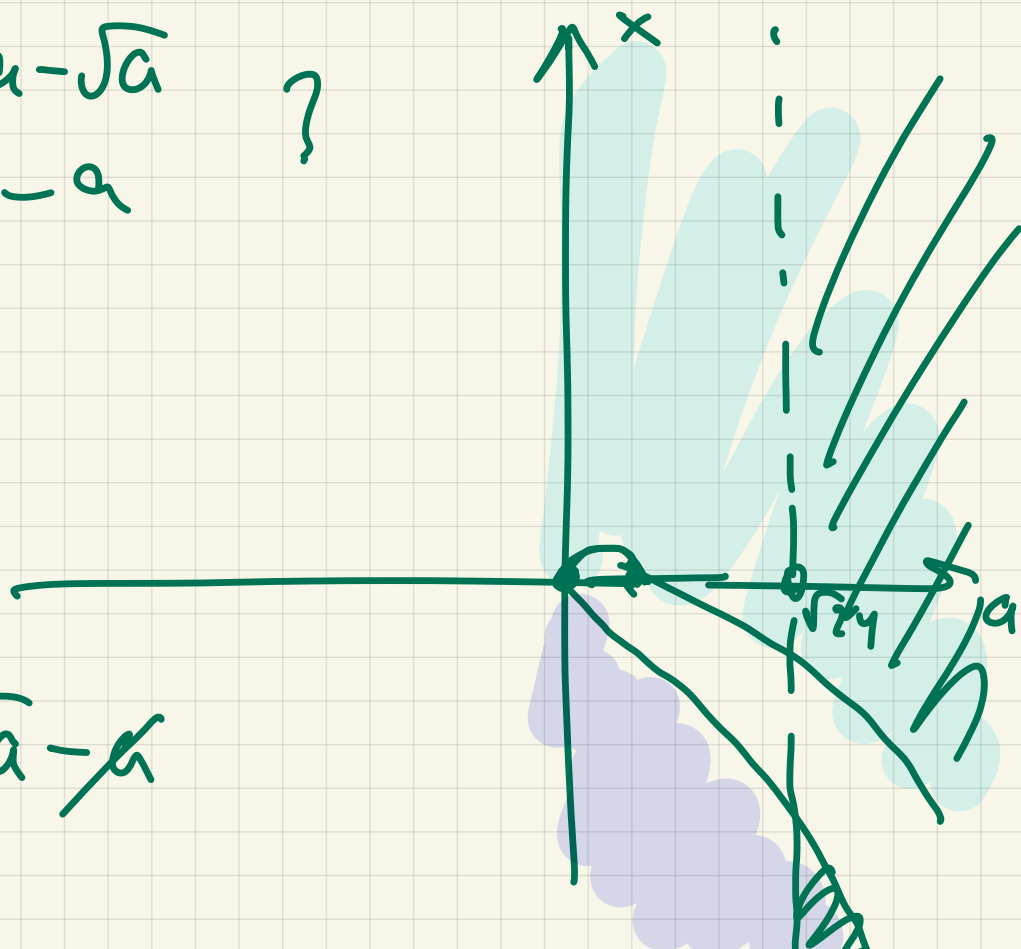
$$x^2 + 2ax + a^2 = (x+a)^2 - a \leq 0$$

$$(x+a+\sqrt{a})(x+a-\sqrt{a}) \geq 0$$

$$x = -a - \sqrt{a}$$

$$x = \sqrt{a} - a$$

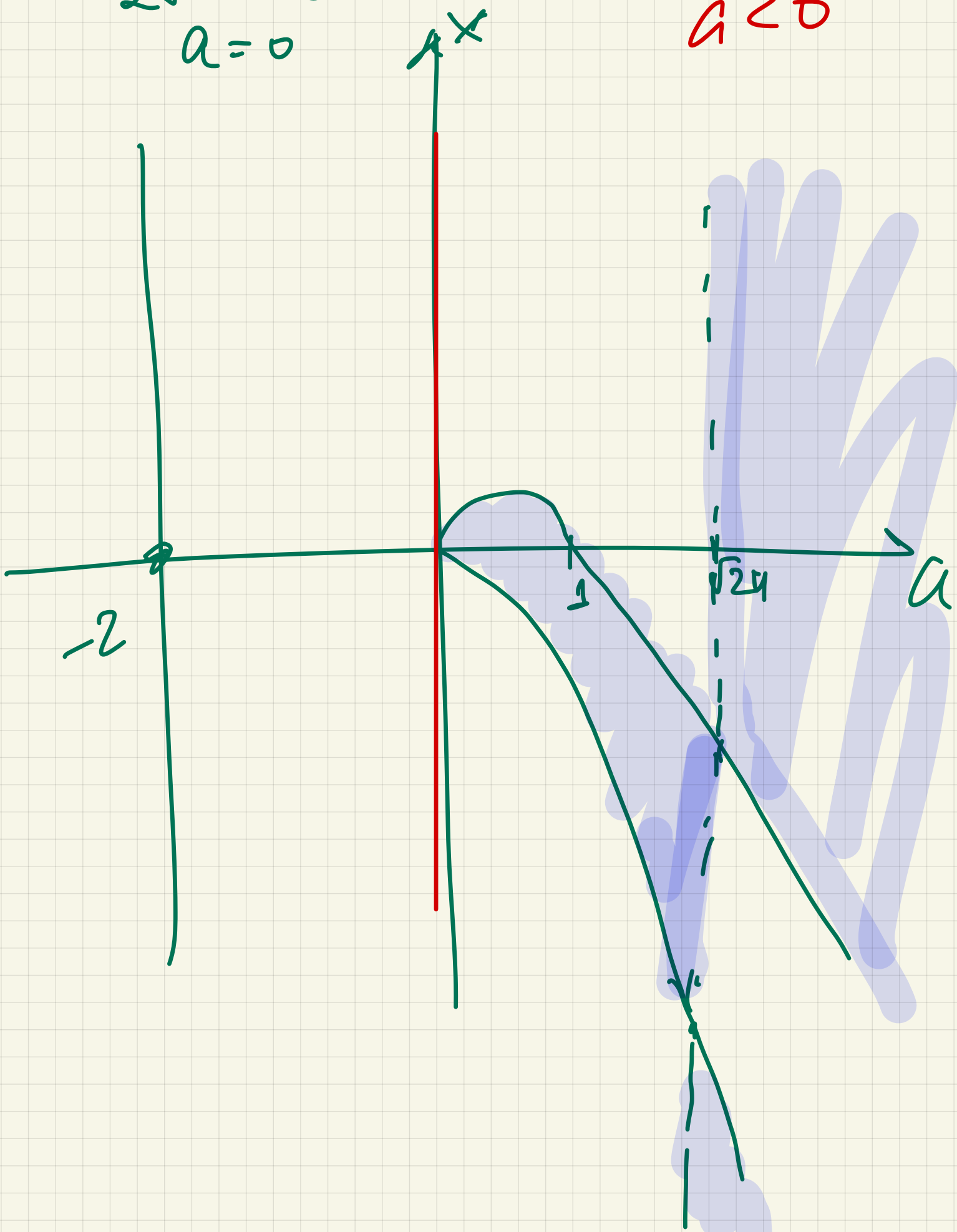
a	$\frac{1}{4}$	$\frac{1}{64}$	$\frac{1}{36}$
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$$-a - \sqrt{a} = \sqrt{a} - a$$

$$2\sqrt{a} = 0$$
$$a = 0$$

$$A < 0$$



$$|x+3a+10| + |x-3a-14| \leq 3|x| + 3|x-2|$$

$\forall x ?$

$$3a+10+x$$

$$x-3a-14$$

$$x$$

$$x-2$$

$$a = \frac{-x-10}{3}$$

$$a = \frac{x-14}{3}$$

